Project 2: Real-Time IoT Device Monitoring & Cloud Dashboarding Platform

Who am I ?

2015: SupCom Tunis

2015 - 2018: Master HES-SO (MSE)

2019 - …: SDATAWAY

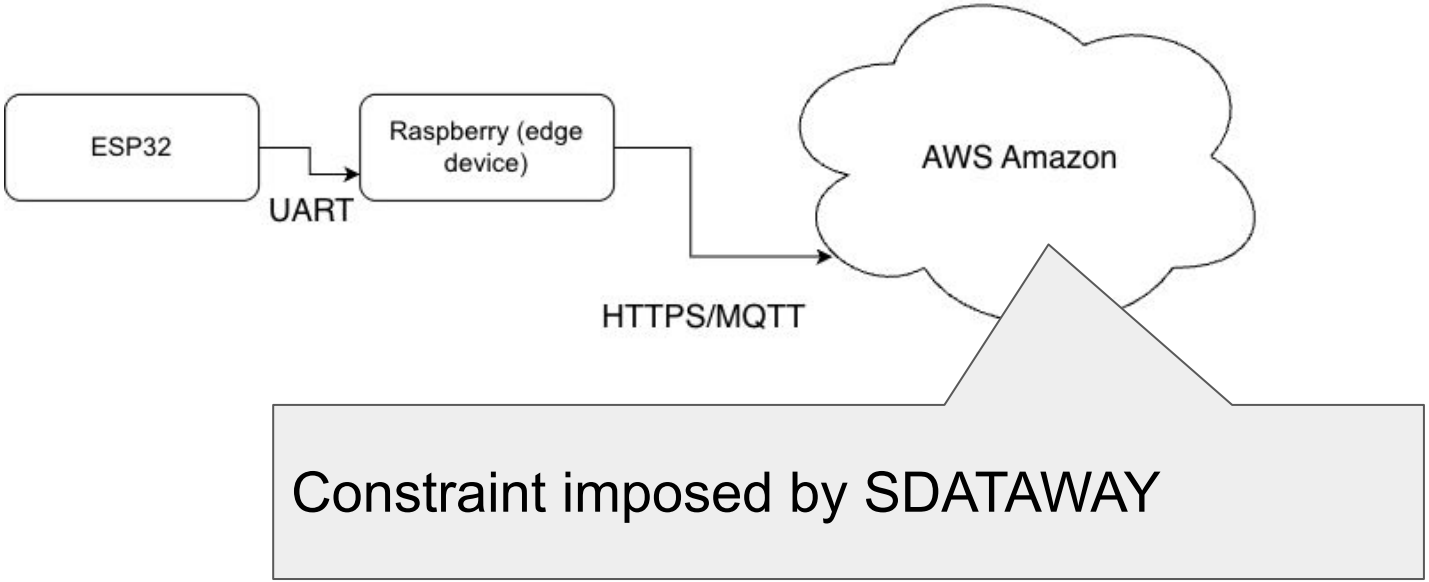


Nizar Bouchedakh

Project 2: Real-Time IoT Device Monitoring & Cloud Dashboarding Platform

- Client : https://www.sdataway.com/
- Key contact: Nabil Abdennadher, Raoul Dupuis, Nizar Bouchedakh (SDATAWAY)
- Ojectives:
  - Simulate monitoring data for a smart capsule coffee machine and streams it as CSV over a serial port (or stdout)
  - Design and implement a data pipeline that pushes real-time device monitoring data to the cloud.
  - Design and implement a dashboarding/analytics solution to visualize and monitor this data.

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How to simulate the ESP32 ?

- The coffee machine uses a MCU (ESP32)
- We provide you with a software simulator. Thanks Nizar!
- The simulator is available on Teams.
- You don’t need to go through its code, just setup config.yaml

Why edge device ?

- Data is generated at a high sampling rates
  - We need edge device with more memory and processing power
  - SDATAWAY wants to avoid integrating the monitoring code into the firmware.

Scenarios

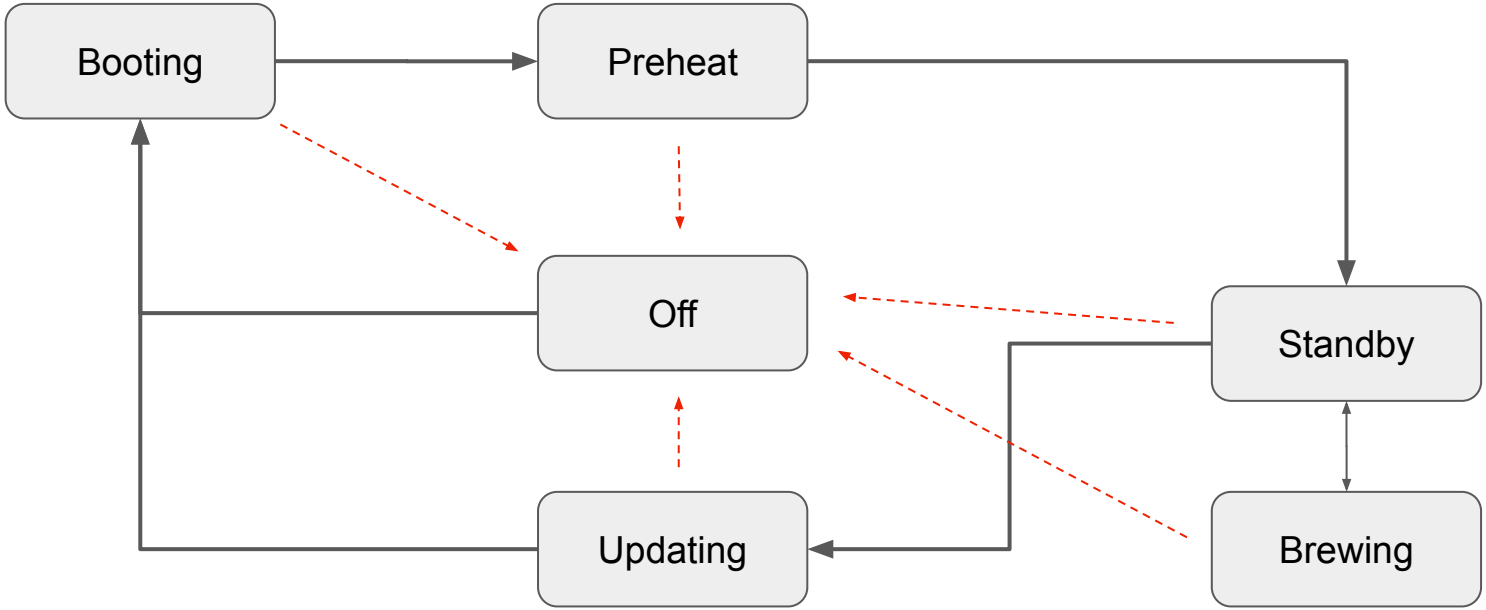
- Themal_runway
- Memory_leak_fota
- Out_of_memory
- Silent_device

See details in the README.md on Teams
(coffee-machine-simulator-Project2.zip)

Status device

- **Off**: appliance switched off (heater off); the control board stays, powered and keeps reporting.
- **Booting**: control board (re)starting, e.g. after power-on or a reboot.
- **Preheat**: heating from ambient up to the brew setpoint.
- **Standby**: idle, at temperature, ready to brew.
- **Brewing**: actively brewing (pump + heater engaged).
- **Updating**: a simulated firmware update is in progress
- **Degraded**: early warning (approaching an overheat or low-memory, threshold); overrides the lifecycle state above.
- **Error**: an active fault, with `error_code` bit(s) set; overrides, everything else.

The status device



For sake of simplicity, the two status : *Degraded* and *Error* are not shown in this slide.

Next steps

- Develop and deploy the edge module: Several IoT devices are assumed to be connected to the same edge device
- Define the communication between the edge and the cloud (AWS)
- Visualise the data
- Cloud services to identify the scenarios (patterns)
- Scalability and cost

Next week

- Architecture of the platform: what cloud services to use for each component of the whole application ?
- The Edge module deployed on a VM (to simulate an edge device)
- The edge module is connected to at least 5 IoT devices (use the simulator to simulate the IoT devices)